

NGHIA NGUYEN-TRUNG

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EDUCATION

University of North Carolina at Chapel Hill · Chapel Hill, NC, USA

- **Ph.D. in Statistics and Operations Research** Aug 2022 – May 2027 (Expected)
 - Status: Advanced Ph.D. Candidate (Year 5).
 - Ph.D. Advisor: Prof. Quoc Tran-Dinh.
 - Dissertation: *Advanced Fast First-Order Methods for Generalized Equations: Randomized, Inexact, and Preconditioned Techniques.*
- **M.S. in Statistics, Analytics and Data Science** Aug 2022 – Dec 2025

Hanoi University of Science and Technology · Hanoi, Vietnam

- **Engineer in Mathematics and Informatics** Aug 2017 – May 2022
(Equivalent to B.S. + 1 year of advanced graduate coursework)
 - **GPA:** 3.70/4.00 – **Honors:** Valedictorian of the Class of 2022 (Ranked 1st).
 - **Distinction:** Thesis-based Degree with Highest Honors.

PROFESSIONAL MEMBERSHIPS

- **Student Member**, Society for Industrial and Applied Mathematics (SIAM).
- **Student Member**, Institute for Operations Research and the Management Sciences (INFORMS).

RESEARCH EXPERIENCE

University of North Carolina at Chapel Hill · Graduate Research Assistant Aug 2023 – Present

- Developed a unified framework for deterministic extragradient and splitting methods, integrating Halpern anchoring and Nesterov acceleration to solve non-monotone operator inclusions. Established optimal $\mathcal{O}(1/k)$ and $o(1/k)$ last-iterate convergence guarantees on mapping residual norms under co-hypomonotonicity.
- Developed novel variance-reduced algorithms designed for non-monotone operator inclusions and large-scale saddle-point problems, using both unbiased and biased oracles. Established rigorous sublinear convergence rates for stochastic composite optimization with application to adversarial learning.
- Engineered accelerated fixed-point algorithmic frameworks utilizing delayed inexact oracles. Optimized high-dimensional machine learning workflows for parallel and high-performance computing architectures.
- Formulated decentralized first-order optimization routines for composite monotone inclusions over distributed communication networks.

Institute of Mathematics, Vietnam Academy of Science and Technology · Visiting Researcher Feb 2022 – Jul 2022

- Investigated projection-based numerical methods for split variational inequalities and generalized equilibrium problems. Applied theoretical bounds to solve constrained optimal control problems.

Institute of Mathematics, VAST · Research Intern Apr 2021 – Jul 2021

- Conducted research on semidefinite programming relaxations for extremal problems in graph theory and combinatorics.

Hanoi University of Science and Technology · Undergraduate Research Assistant Aug 2020 – Jul 2022

- Formulated novel self-adaptive and inertial-type iterative algorithms for multiple-set split variational inequalities (MSVIP) and split feasibility problems with multiple output sets (SFP MOS). Eliminated requirements for global operator norm estimations, Lipschitz constants, or line searches by designing parameter-free dynamic step-size criteria.
- Extended splitting frameworks to quasimonotone bilevel variational inequalities and level-set constrained optimization via alternated inertial extrapolation and ball-relaxed projections. Proved rigorous strong convergence theorems in real Hilbert spaces under mild structural assumptions.
- Implemented parallel computing frameworks for generalized split feasibility problems to enhance computational efficiency. Demonstrated practical utility by applying proposed algorithms to discrete linear optimal control problems and high-dimensional sparse recovery models, including LASSO with Elastic Net regularization.

Preprints and Submitted Manuscripts

1. **Nghia Nguyen-Trung**, Quoc Tran-Dinh. *A Class of Accelerated Fixed-Point-Based Methods with Delayed Inexact Oracles and Its Applications*. arXiv preprint arXiv:2512.13547, 2025.
2. Quoc Tran-Dinh, **Nghia Nguyen-Trung**. *Unbiased and Biased Variance-Reduced Forward-Reflected-Backward Splitting Methods for Stochastic Composite Inclusions*. arXiv preprint arXiv:2603.15576, 2026 (Submitted to **Computational Optimization and Applications**).
3. Quoc Tran-Dinh, **Nghia Nguyen-Trung**. *VFOG: Variance-Reduced Fast Optimistic Gradient Methods for a Class of Nonmonotone Generalized Equations*. arXiv preprint arXiv:2508.16791, 2025 (Submitted to **Mathematics of Operations Research**).
4. Quoc Tran-Dinh, **Nghia Nguyen-Trung**. *Accelerated Extragradient-Type Methods – Part 2: Generalization and Sublinear Convergence Rates under Co-Hypomonotonicity*. arXiv preprint arXiv:2501.04585, 2025 (Submitted to **Computational Optimization and Applications**).

Peer-Reviewed Journal Articles

1. Thuy Nguyen Thi Thu, **Nghia Nguyen-Trung**. *A Strongly Convergent Theorem for a Class of Quasimonotone Bilevel Split Variational Inequality Problems*. Vietnam Journal of Mathematics, 2026. DOI: [10.1007/s10013-026-00816-8](https://doi.org/10.1007/s10013-026-00816-8).
2. Quoc Tran-Dinh, **Nghia Nguyen-Trung**. *Revisiting Extragradient-Type Methods – Part 1: Generalizations and Sublinear Convergence Rates*. Computational Optimization and Applications, 2026. DOI: [10.1007/s10589-025-00727-3](https://doi.org/10.1007/s10589-025-00727-3).
3. Thuy Nguyen Thi Thu, **Nghia Nguyen Trung**. *An inertial-type algorithm for a class of bilevel variational inequalities with the split variational inequality problem constraints*. Optimization, 74(3), 589–613, 2025. DOI: [10.1080/02331934.2023.2262493](https://doi.org/10.1080/02331934.2023.2262493).
4. Thuy Nguyen Thi Thu, **Nghia Nguyen Trung**. *Some novel inertial ball-relaxed CQ algorithms for solving the split feasibility problem with multiple output sets*. Journal of Applied Analysis and Computation, 14(3), 1485–1507, 2024. DOI: [10.111948/20230259](https://doi.org/10.111948/20230259).
5. Thuy Nguyen Thi Thu, **Nghia Nguyen Trung**. *A hybrid projection method for solving the multiple-sets split feasibility problem*. Computational and Applied Mathematics, 42, Art. 292, 2023. DOI: [10.1007/s40314-023-02416-5](https://doi.org/10.1007/s40314-023-02416-5).
6. Thuy Nguyen Thi Thu, **Nghia Nguyen Trung**. *A new iterative method for solving the multiple-set split variational inequality problem in Hilbert spaces*. Optimization, 72(6), 1549–1575, 2023. DOI: [10.1080/02331934.2022.2031193](https://doi.org/10.1080/02331934.2022.2031193).
7. Thuy Nguyen Thi Thu, **Nghia Nguyen Trung**. *A parallel algorithm for generalized multiple-set split feasibility with application to optimal control problems*. Taiwanese Journal of Mathematics, 26(5), 1069–1092, 2022. DOI: [10.11650/tjm/220502](https://doi.org/10.11650/tjm/220502).

Working Papers in Preparation

1. **Nghia Nguyen-Trung**, Ion Necoara, Quoc Tran-Dinh. *Decentralized Fast Fixed-Point Algorithms for Composite Monotone Inclusions*. In preparation, 2026.
2. **Nghia Nguyen-Trung**, Ion Necoara, Quoc Tran-Dinh. *Optimistic Gradient Algorithms with Randomized Coordinate and Variance Reduction Strategies for Composite Inclusions*. In preparation, 2026.

CONFERENCE PRESENTATIONS & TALKS

1. **[Target SIAM Presentation]** “A Class of Accelerated Fixed-Point-Based Methods with Delayed Inexact Oracles and Its Applications”, *SIAM Conference on Mathematics of Data Science (MDS26)* (Poster Presentation), Nov 2026.
2. “Optimistic Gradient Algorithms with Randomized Coordinate and Variance Reduction Strategies for Composite Inclusions”, *INFORMS Annual Meeting* (Contributed Talk, Scheduled), Nov 2026.
3. “Optimistic Gradient Algorithms with Randomized Coordinate and Variance Reduction Strategies for Generalized Equations”, *Graduate Student Seminar*, STOR Department, UNC–Chapel Hill, Oct 31, 2025.
4. “New Hybrid Projection Method and Shrinking Projection Method for Solving Multiple-Set Split Variational Inequalities”, *20th Workshop on Optimization and Scientific Computing*, Hanoi, Vietnam, Apr 21–23, 2022.
5. “A New Hybrid Projection–Proximal Point Algorithm for Split Variational Inequality Problems”, *VIASM International Conference on New Trends in Numerical Optimization and Applications*, Hanoi, Vietnam, Dec 26–27, 2021.
6. “A New Self-Adaptive Algorithm for Solving Multiple-Set Split Variational Inequalities”, *Conference on Contemporary Issues in Mathematics and Applications*, VNU, Hanoi, Vietnam, Oct 30–31, 2021.
7. “Semidefinite Programming and Applications in Combinatorial Optimization”, *Institute of Mathematics, VAST*, Hanoi, Vietnam, Jul 2021.
8. “Strong Convergence Analysis for Split Feasibility Problems in Hilbert Spaces”, *19th Workshop on Optimization and Scientific Computing*, Hanoi, Vietnam, Apr 22–24, 2021.

HONORS AND AWARDS

- Silver Reviewer Award** · ICML 2026 2026
- Awarded to top-performing reviewers at the 43rd International Conference on Machine Learning based on Area Chair evaluation.
- Excellence in Teaching Assistance and Instruction Award** · UNC–Chapel Hill 2024, 2025
- Recognized by the department for exceptional undergraduate teaching and leadership as Instructor of Record.
- Valedictorian of the Class of 2022** · Hanoi University of Science and Technology 2022
- Ranked 1st overall among all graduating students in the School of Applied Mathematics and Informatics.
- Scientific Research Contest – Second Prize** · HUST 2021
- Recognized for original algorithmic development in split feasibility problems, later published in *Optimization*.
- National Mathematics Development Program Fellowship** · VIASM 2020 – 2021
- Highly competitive national scholarship awarded by the Vietnam Institute for Advanced Study in Mathematics.
- HUST Academic Excellence Scholarship (Type A)** · HUST 2018 – 2022
- Maintained top 1% merit ranking across all undergraduate semesters.

PROFESSIONAL SERVICE

- Journal & Conference Peer Reviewer:**
- Journal of Optimization Theory and Applications (JOTA) – Springer
 - International Conference on Machine Learning (ICML)
 - Neural Information Processing Systems (NeurIPS)
 - EURO Journal on Computational Optimization – Elsevier
 - Optimization and Engineering – Springer
 - Journal of Computational and Applied Mathematics – Elsevier

TEACHING EXPERIENCE

- Graduate Teaching Fellow (Instructor of Record)** · UNC–Chapel Hill May 2024 – Present
- Fully responsible for course design, lectures, and examination for STOR 113 (*Decision Models for Business and Economics*) and STOR 155 (*Introduction to Data Models and Inference*).
- Graduate Teaching Assistant** · UNC–Chapel Hill Aug 2022 – Present
- Led recitations, office hours, and grading for Ph.D.-level and advanced undergraduate optimization courses: STOR 612: *Foundations of Optimization* (Ph.D. level), STOR 512: *Optimization for Machine Learning*, and STOR 415: *Introduction to Optimization*.

TECHNICAL EXPERTISE

- Languages & Computing:** Python (PyTorch, NumPy, SciPy, Pandas), MATLAB, R, $\LaTeX$.
- Optimization Solvers:** Gurobi, CvxPy, CvxOpt, AMPL, GAMS.
- Methodologies:** First-Order Methods, Variance Reduction, Stochastic Optimization, Asynchronous and Parallel, Decentralized and Consensus.